

MARAPPA REDDY CHURNIKA

AI/ML ENGINEERING | DATA ENGINEERING | SOFTWARE ENGINEERING

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Professional Summary

M.S. Data Science candidate building production-style **AI/ML, Generative AI, data engineering, and full-stack systems**. Strong in **Python, SQL, PyTorch, RAG, LangGraph, FastAPI, Spark, Airflow, PostgreSQL, REST APIs, Docker, and CI/CD**, with experience in data pipelines, model evaluation, vector search, testing, and deployment.

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, MATLAB

AI/ML & GenAI: PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost, Machine Learning, Deep Learning, Generative AI, RAG, LangGraph, LangChain, Embeddings, Vector Search, Cross-Encoder Reranking, MLflow

Data & Backend: Spark, Airflow, Pandas, NumPy, ETL/ELT, Data Pipelines, Feature Engineering, Data Quality, FastAPI, PostgreSQL, Qdrant, Redis, REST APIs, FHIR R4

Frontend / DevOps / Cloud: React, Next.js, Streamlit, Docker, Docker Compose, Git, GitHub Actions, CI/CD, AWS S3, CloudWatch, Render, Vercel

Experience

University at Buffalo, SUNY

Mar 2026 – May 2026

Research Assistant – Generative AI / Machine Learning

Buffalo, NY

- Developed and evaluated a few-shot handwriting style-transfer **diffusion workflow** using PyTorch, latent U-Net, DDIM, Stable Diffusion VAE, CANINE text embeddings, and MobileNetV2 style embeddings.
- Engineered IAM/GNHNK preprocessing, writer mappings, data loaders, cached tensors, training/EMA checkpoints, validation, and reproducibility; analyzed unseen-writer generalization and failure modes, compared a VAE + autoregressive Transformer baseline, and built an interactive generation/validation/reranking application.

Tirumala Tirupati Devasthanams (TTD)

Jun 2024 – Aug 2024

Web Development Intern

Tirupati, India

- Contributed to frontend, backend, database, and **REST API** workflows; performed software/API testing, debugging, defect reproduction, data-flow validation, and technical documentation.

Slash Mark

Oct 2023 – Feb 2024

Python Intern

Remote

- Developed **Python automation and data-processing workflows** with validation, exception handling, debugging, and output verification to improve reliability and maintainability.

Selected Projects

CareFlow AI – Agentic Healthcare Intelligence & RAG Platform

Live Demo | GitHub

- Architected and deployed an end-to-end **Agentic AI / RAG platform** combining public CMS Medicare policy retrieval with synthetic **FHIR R4** and CMS DE-SynPUF workflows using **Python, FastAPI, LangGraph, PostgreSQL, Qdrant, Redis, and Next.js**.
- Engineered **BM25 + dense retrieval, RRF, cross-encoder reranking, citation validation, abstention, bounded multi-agent orchestration, HITL review, audit trails, caching, analytics, unit/integration/E2E testing, and GitHub Actions CI/CD**; dev retrieval achieved **0.96 Hit@1** and **0.98 MRR@5**.

Citi Bike Demand Prediction – End-to-End Data & ML Platform

GitHub

- Built an end-to-end **ETL/ML pipeline** using Airflow, Spark, Iceberg, PostgreSQL, FastAPI, Streamlit, and Docker with SHA-256 ingestion tracking, WAP validation, quarantine/deduplication, and **Bronze/Silver/Gold** layers.
- Engineered lag, rolling, trend, and cyclical features; trained Logistic Regression, Random Forest, and LightGBM with Optuna, reducing cumulative operational penalty by **50.9% versus a naive baseline** on a selected station/window.

AI-Powered SQL Query Assistant

Python | PostgreSQL | OpenAI API | LangChain | Streamlit

- Built a **natural-language-to-SQL** assistant with LLM-based query generation, SQL validation, PostgreSQL execution, authentication, and an interactive Streamlit interface.

Medical IoT Security – ML Cyberattack Detection

Python | CNN | Isolation Forest | CICIoMT2024

- Developed Isolation Forest and CNN-based detection across **18 cyberattack categories**; evaluated precision, recall, F1, specificity, NPV, ROC-AUC, confusion matrices, and false-positive/false-negative behavior.

Education

University at Buffalo, State University of New York (SUNY)

Expected Dec 2026

M.S. Engineering Science – Data Science | **GPA: 3.704/4.000**

Buffalo, NY

Relevant Coursework: Statistical Learning I & II, Machine Learning, Data Intensive Computing, Programming & Database Fundamentals, Data Models & Query Languages, Probability Theory, End-to-End Machine Learning

Amrita Vishwa Vidyapeetham

May 2025

B.Tech – Computer Science & Engineering

Coimbatore, India

Leadership: ANOKHA TechFair Head / Overall Coordinator (2023–2024) | Class Representative | NSS Volunteer (2021–2023)